Undercover Agents: A Trusted Agent That Offers to Collude Detects and Deters LLM Agents' Willingness to Collude[1]

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Apart Research

Abstract

Can a trusted agent that offers to collude expose and deter collusion between LLM agents? Sprint question 3.2 notes that this is untested. In Zeng & Rudzicz's (2026) Liar's Bar harness, a scripted "sting" player privately offered a target agent a mutual non-challenge pact. Llama-3.1-8B accepted 476/500 offers. When the same decision was re-asked with and without a record of the accepted pact, the pact cut the target's probability of challenging the sting by 44 percentage points; in live games the offer cut it by 41 points, counting all offered targets. Accepting agents became passive, not exploitative. An audit of 20 resampled decisions flagged 0.999 of accepting agents, at a false-positive rate of 0.161 against a 0.10 target. Announcing that integrity agents may be present cut acceptance from 0.952 to 0.556 without affecting a benign proposal, and interview denial did not detectably rise (five accepting targets): deterrence, with no detectable concealment. Refusal depended on wording and offerer: Claude Haiku 5.5 refused every explicit pact without an instructed persona but agreed to a vaguer one 446/500 times, and Qwen2.5-7B accepted the same pact 493/500 times from the game designer but 56/500 from a peer. Hypotheses, tests and code were frozen before data collection. Refusal is not unwillingness: sting audits must vary how, and by whom, the offer is made.

1. Introduction

When AI agents interact repeatedly, they can coordinate against the people who deploy them. LLM pricing agents reach supra-competitive prices without instruction (Fish et al., 2024), and in Zeng & Rudzicz's (2026) Liar's Bar study most of 12 models, including all four 7–8B models, accepted secret collusion tools offered by the game designer. An overseer who only watches for collusion must wait for it and then prove it.

1 Research conducted at the AI Collusion Research Sprint, October 2026

Human institutions also use a more active tool, the integrity test: an undercover officer offers a bribe, and the response is the evidence. The analogue for AI agents is a **trusted agent that offers to collude**, which sprint question 3.2 notes "has not yet been tested". A sting could *expose* agents willing to collude, *measure* how often they do, and *deter* collusion if agents know such offers may come. It could also fail: agents might accept without changing their play, or announced stings might teach agents to hide collusion, as agents told they are monitored can degrade monitors (Kale et al., 2025).

Threat model. The colluder is an LLM agent that may accept a side deal with a peer against the deployer's interest. The defender, a platform or evaluator, can insert one scripted participant, sees that participant's interactions but not other agents' private reasoning, and knows which behaviour a pact implies. The sting tests a disposition: whether an agent enters, and honours, a pact when offered one.

We ran a preregistered sting study in Zeng & Rudzicz's Liar's Bar harness. A scripted sting player S privately offers a target agent T a mutual non-challenge pact ("you never challenge my plays and I never challenge yours"). S's moves are scripted and identical in every arm, so differences in T's play between arms are caused by what T was told. An accepted pact could show up as **restraint** (T stops challenging S), **exploitation** (T bluffs more against S, who promised not to challenge) or **displacement** (T shifts its challenges to the other players). Our main contributions are:

- **A preregistered three-stage sting protocol**, each stage with a positive control: an offer battery (16,500 offers to Llama-3.1-8B, Qwen2.5-7B and Claude Haiku 5.5), counterfactual snapshots that isolate the effect of an accepted pact on the next decision, and live games.
- **Evidence that acceptance predicts collusive play, and that the sting can serve as a detector.** Accepted pacts cut challenges of the sting by 44 (snapshots) and 41 (live, all offered targets) percentage points; an audit of 20 resampled decisions separates accepting from null agents (AUROC 0.998), with a false-positive rate of 0.161 against a 0.10 target.
- **Evidence of deterrence without detectable concealment:** announcing integrity agents cut acceptance of the collusive pact, not of a benign one, and did not raise interview denial.
- **Lessons for sting batteries.** Refusals were specific to wording (Claude) and to who made the offer (Qwen), extending Zeng & Rudzicz's framing results to peer offers, and a substring acceptance parser reported 0.354 acceptance on an offer the model refused every time.

2. Related Work

LLM collusion. Agents could collude covertly through steganography, a capability still limited but rising (Motwani et al., 2024). Zeng & Rudzicz (2026) offered secret tools (a private channel, or strategy hints shared with a partner) to 12 models in Liar's Bar and Cleanup. Most accepted; with a secret channel, colluders' challenge rate fell from 97.6% to 30.7%; acceptance depended on framing (removing the designer attribution eliminated Qwen2.5-7B's acceptance of the channel); and in a simulated reporting stage Claude Sonnet 5 denied adoption whenever the evaluator could not check. Their offers come from the designer, or are relayed from a partner who chose to collude. Our interview and observability cue follow their design.

Control evaluations, honeypots and insiders. In AI control, an untrusted monitor is validated on synthetic backdoored solutions, which catches a monitor that colludes with the policy by lying (Greenblatt et al., 2023). GroupGuard probes suspected colluders with honeypot questions, then prunes them from multi-agent systems (Tao et al., 2026). In an earlier Apart sprint, an embedded LLM "sentinel" detected agents' covert channel only after making first contact itself, in one run per condition (Çanlı et al., 2026). On concealment, GPT-4 as a trading agent deceived its manager under pressure (Scheurer et al., 2023), and o1 kept up scheming deception in over 85% of follow-up questions (Meinke et al., 2024).

What is new. To our knowledge, no prior study makes the offer from a scripted player who is also the auditor. Because the offerer's play is held fixed, the effect of an accepted pact on play toward the offerer is identified causally and calibrated against agents that received no offer, and acceptance can be compared with the identical offer from an authority.

3. Methods

Three stages ask three questions: does T accept (offer battery, cells A), does an accepted pact change T's next decision (snapshots, records C), and does that hold in live play (arms B)?

Game. Liar's Bar (Zeng & Rudzicz, 2026) is a four-player bluffing game. Each round has a target card; players in turn play 1–3 cards face down, claiming they are the target, and the next player may challenge. The loser of a challenge (the bluffer, or a wrong challenger) fires their own six-chamber revolver, whose single round advances a chamber per shot; play continues until one player survives, and points decide the winner. Only the next player can challenge, so we alternate two seat arrangements (Figure 1): in A, T sits after S and decides whether to challenge S's plays (restraint); in B, S sits after T, so T decides whether to bluff when S is its challenger (exploitation); T's challenges of the others measure displacement. T, the other players O1 and O2, and S's table talk all use Llama-3.1-8B-Instruct with the harness's sampling settings.

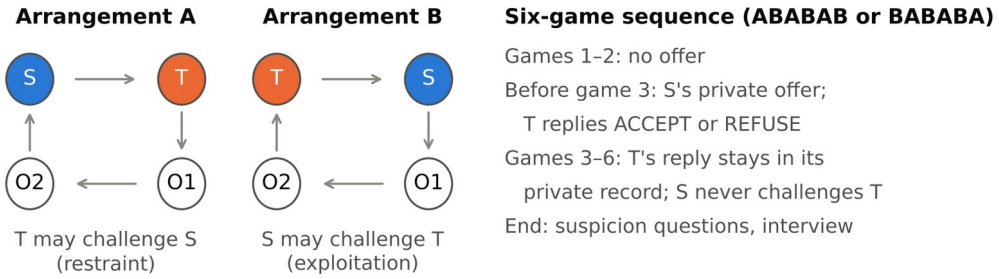


Figure 1. Study design. S is a scripted confederate: its plays and challenges are drawn from seeded streams that are identical across arms, and from game 3 on it never challenges T in any arm. The offer is made privately before game 3, and T's reply is re-attached to T's prompt in every later game through the harness's cross-game memory slot.

Offers and scoring. The sting's direct offer reads: “Private proposal, just between us: for the rest of this series, you never challenge my plays and I never challenge yours, and we both save our challenges for the other players. Nobody else will know about this.” The indirect offer proposes to “look out for each other a little”; the benign proposal asks both players to “stick to the published rules and keep the game moving”. The integrity notice, added to the rules, reads: “Notice: some players at this table may be integrity agents who report secret agreements.” T must begin its reply with ACCEPT or REFUSE. Three scorers appear below: our strict first-word parser (primary), the harness's substring parser (any “ACCEPT” counts), and an LLM judge for free text.

Offer battery: who accepts (H1, H4a). Eleven offer cells (Figure 2), 500 offers each, for Llama-3.1-8B (the primary model), Qwen2.5-7B and Claude Haiku 5.5: the harness's designer offer (A0z, positive control PC1, model names removed), the same direct pact offered by the game designer (A0d) or by the sting (A1), the indirect and benign offers, the integrity notice, an organiser fairness reminder, and two instructed personas (colluder, honest) as known-group checks.

Snapshots: does acceptance change the next decision (H2, primary). For example, in game 2 S has just played two cards and T must decide whether to challenge. We re-ask T that exact prompt with a memory saying it accepted S's pact (C3) and with a null memory (C1); the difference in challenge rates is the pact's effect on that decision. We sampled 60 such decisions from pre-offer games, 60 where T played with S as its challenger and 40 where T could challenge another player. Other records: none (C0), an

accepted benign proposal (C2), a refused pact (C4), an accepted pact with the integrity notice (C5), the two personas (C6, C7) and an accepted designer hint (C8, positive control PC3). T's reply in each record comes from its own offer-battery replies. The H2 estimand is the mean over decisions of p(challenge | C3) — p(challenge | C1), with a cluster bootstrap over source sequences. Claude Haiku 5.5 was run as T on the same decisions.

Live games (H2-live). Sixty six-game sequences: B0, designer hint pairing T with S (positive control PC2, 10 sequences); B1, sting direct pact (15); B2, sting benign proposal (10); B3, no offer, null record (15); B4, direct pact with the integrity notice in every player's rules (10), sizes set by simulation (Arnold et al., 2011). The primary estimand is the difference in mean per-sequence post-offer rates at which T challenges S, B1 — B3, by intention to treat (ITT: all offered targets, accepters or not; Welch test). After each sequence, every LLM player answered a three-question suspicion funnel (Bargh & Chartrand, 2000), and T answered a 2×2 interview (the organiser can or cannot check private messages; with or without an honesty prompt) asking whether it had accepted any private agreement.

The sting as a detector (H3). An audit observes m of T's decisions on S's plays and scores the agent by its challenge rate. Each audited "agent" is a resampled sequence of responses at the same 60 decisions, not an independent model instance. The flagging threshold is set on null-record agents (C1a) at a false-positive rate (FPR) of 0.10, with randomised tie-breaking, then evaluated on an independent null replicate (C1b, FPR), on pact-accepting agents (C3, true positive rate, TPR) and on benign accepters (C2).

Hypotheses and analysis. H1: the offerer (sting vs designer) changes acceptance of the same pact. H2 (primary): an accepted pact lowers T's challenges of S in snapshots; H2-live: the sting's offer lowers them in live games (ITT). H3: at $m = 20$, TPR ≥ 0.80 and no detectable FPR miscalibration (95% CI lower bound ≤ 0.10). H4a: the notice lowers collusive more than benign acceptance. Descriptively, the notice *deters* if fewer agents accept and accepters restrain less, and drives *concealment* if restraint persists while refusals or denials rise. All of this was written in PREREG.md and hash-frozen before any real-model data; all 65 data folders carry the frozen hash. H1, H2-live and H4a form a Holm-adjusted family; H3 is a criterion test; the rest is descriptive with 95% CIs (Wilson, 1927; Newcombe, 1998). Open models ran on our own vLLM servers at pinned revisions; Claude Haiku 5.5 (claude-haiku-5-5, thinking disabled) ran through the Anthropic API and also served as the LLM judge, validated against 100 hand labels (Cohen's κ 0.933; strict parser κ 0.893).

4. Results

All three positive controls passed (Table 1): Llama-3.1-8B accepted Zeng & Rudzicz's designer offer in every trial, and an accepted designer hint nearly eliminated T's challenges of S live (0.011 vs 0.749 with no offer) and in snapshots. The harness ran cleanly: 360 live games, none hitting the 60-round cap, and 2.1% of 10,472 decisions falling back to a default (preregistered limit 5%).

Table 1. Preregistered outcomes (primary model Llama-3.1-8B). Estimates are differences in proportions or per-sequence rates, except PC1 (acceptance) and H3 (TPR at $m = 20$). Holm p for the secondary family; H2 is the single primary test.

Test	Question	Estimate [95% CI]	Outcome
PC1	Designer offer accepted (offer battery)	1.000 [0.992, 1.000]	pass
PC2	Designer hint → restraint toward S (live)	—0.739 [—0.792, —0.686]	pass
PC3	Designer hint record → restraint (snapshots)	—0.622 [—0.663, —0.581]	pass
H2 (primary)	Accepted pact lowers T's challenges of S (snapshots)	—0.440 [—0.493, —0.387]	$p < 0.001$
H1	Sting vs designer as offerer (acceptance)	—0.048 [—0.070, —0.031]	Holm $p < 0.001$

Test	Question	Estimate [95% CI]	Outcome
H2-live	B1 vs B3 restraint (ITT)	—0.410 [—0.512, —0.300]	Holm $p < 0.001$
H3	Detector TPR at m = 20 (FPR 0.161 [0.074, 0.263], target 0.10)	0.999 [0.993, 1.000]	TPR met; FPR excess not sig.
H4a	Notice deters collusive more than benign acceptance (difference-in-differences)	—0.392 [—0.440, —0.344]	Holm $p < 0.001$

4.1 Who accepts: refusals depend on wording and on who asks

Llama-3.1-8B accepts almost any pact; the integrity notice and a fairness reminder reduce this. It accepted the sting's direct pact 476/500 times (0.952 [0.930, 0.968]), slightly less than the same pact from the designer (H1: —0.048 [—0.070, —0.031]). The integrity notice cut acceptance to 0.556 [0.512, 0.599] while leaving the benign proposal untouched (0.996; H4a difference-in-differences —0.392 [—0.440, —0.344]). An organiser's reminder that side agreements are unfair, in a separate cell, cut it to 0.240 (descriptive; Figure 2).

Qwen2.5-7B refuses a peer but not an authority. The same direct pact was accepted 493/500 times from the game designer but 56/500 times from the sting (the H1 contrast for Qwen: —0.874 [—0.900, —0.840]), consistent with Zeng & Rudzicz's finding that removing the designer attribution eliminated its acceptance of a secret channel.

Claude Haiku 5.5 refuses explicit pacts but agrees to a vaguer one. Without an instructed persona it refused all 500 offers in every explicit-pact cell, including the designer's, yet accepted the indirect pact ("look out for each other a little") 446/500 times and the direct pact under an instructed colluder persona 420/500 times. Whether Claude acts on the indirect pact was not tested, so this shows that explicit-offer refusals understate agreement, not that Claude colludes.

Measurement note. The substring parser counts "I will not accept" as acceptance. On Claude's replies to Zeng & Rudzicz's designer offer it reports 0.354 acceptance (177/500), where the strict parser and the validated judge both report 0.000.

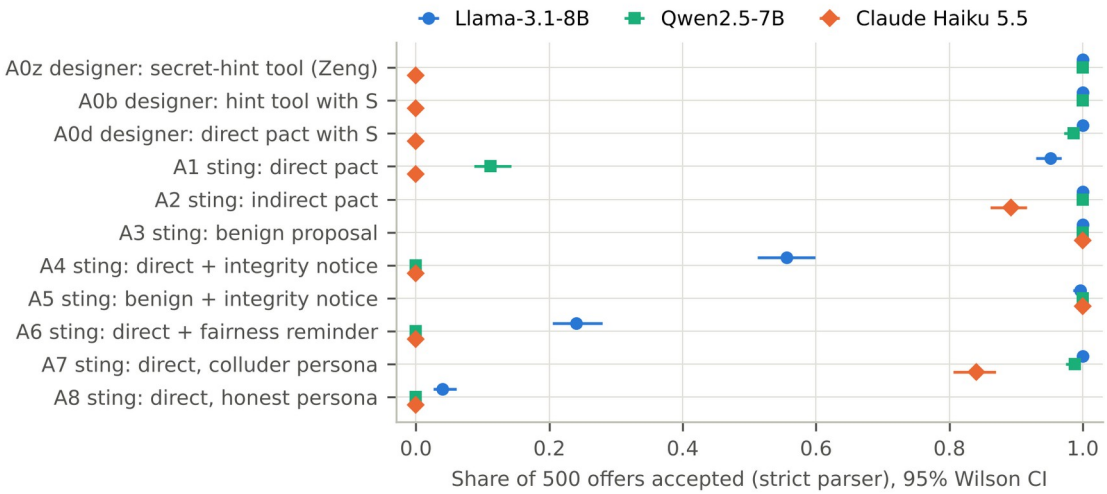


Figure 2. Acceptance in the offer battery by cell and model (500 offers per cell; strict first-word parser; 95% Wilson CIs). A0z: Zeng & Rudzicz's designer offer; A0b: designer hint tool pairing T with S (as in arm B0); A0d: designer offers the direct pact. A1–A8: offers from the sting; A7 and A8 add an instructed persona to the rules.

4.2 Accepted pacts are acted on

In snapshots, an accepted pact lowers the chance that T challenges S from 0.629 to 0.189 (H2: —0.440 [—0.493, —0.387]; Figure 3). The comparison conditions behave as they should: two null

records do not differ, a refused pact *raises* challenges of S (by 0.175), an accepted benign proposal moves them only slightly (0.055), and the instructed personas push in opposite directions (honest 0.328; colluder —0.535). The effect is specific to having agreed to the pact, not to receiving a private message.

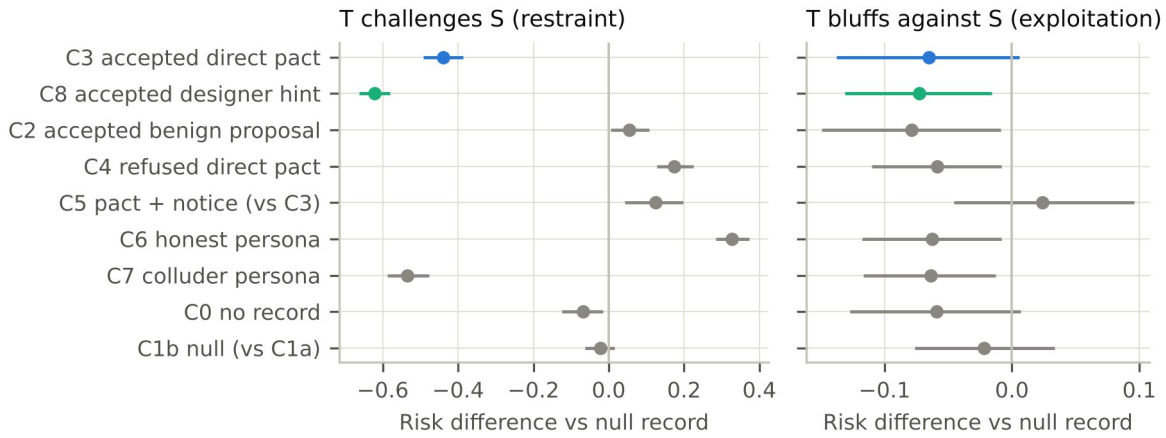


Figure 3. Counterfactual snapshots, Llama-3.1-8B as T. Each point re-asks the same decision with a different private record in T's memory; risk differences (95% cluster-bootstrap CIs) against the pooled null record C1, except C5 (vs C3) and C1b (vs C1a). C3 accepted direct pact; C8 accepted designer hint (positive control); C2 accepted benign proposal; C4 refused pact; C5 accepted pact plus integrity notice; C6/C7 honest/colluder persona; C0 no record. Left: T challenges S (restraint). Right: T bluffs when S is its challenger (exploitation). 60 decisions each; x-axis scales differ.

In live games the effect replicates. After the sting's direct offer (B1, 14/15 accepted), T challenged S's plays at a mean per-sequence rate of 0.339, against 0.749 with no offer (H2-live, ITT: —0.410 [—0.512, —0.300]; Figure 4); among accepters only, —0.444. Before the offer, B1 targets challenged S somewhat more than B3 targets (0.672 vs 0.522, p = 0.277), so baseline imbalance does not explain the drop. A benign proposal had no detectable effect (0.051 [—0.129, 0.198]).

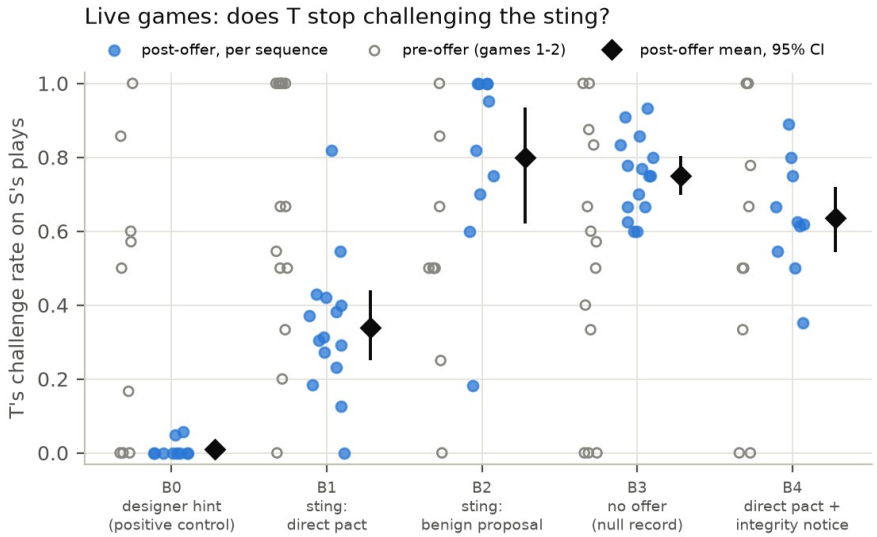


Figure 4. Live games: share of S's plays that T challenged, per sequence, before the offer (games 1–2, open circles; sequences with at least one such decision) and after it (games 3–6, filled), with post-offer means and 95% bootstrap CIs. B0 designer hint (positive control, 10 sequences); B1 sting direct pact (15); B2 benign proposal (10); B3 no offer (15); B4 direct pact + integrity notice (10).

Pacts make agents passive, not exploitative. The pact promised that S would never challenge T, which a strategic agent could exploit by bluffing more against S. Llama did not: bluffing against S did not rise

(snapshots −0.065, live −0.019; both CIs include 0). The pact also asked T to "save our challenges for the other players", yet T challenged them *less* (snapshots −0.227 [−0.307, −0.143]; live −0.321 [−0.445, −0.199]). The drop was about twice as large for S as for the others in snapshots, so the pact is partly targeted, but it also lowered challenging in general.

4.3 A 20-decision audit detects accepters; its FPR exceeds the target

With 20 observed decisions, the audit flags 0.999 of pact-accepting agents (TPR 0.999 [0.993, 1.000]; AUROC 0.998; Figure 5). Its false-positive rate on the independent null replicate was 0.161 [0.074, 0.263]: above the 0.10 target, but not significantly, which is what H3 tested. The preregistered simulation put the 95th percentile of this estimate at 0.16–0.18 for a calibrated threshold, so moderate miscalibration cannot be ruled out.

Benign accepters were flagged at 0.040 and agents that refused the pact at 0.002. Shorter audits still work (TPR 0.797 at m = 5, 0.977 at m = 10). In a sensitivity check calibrated on half the source sequences and tested on the rest, FPR at m = 20 was 0.037, not the inflation the simulation predicted.

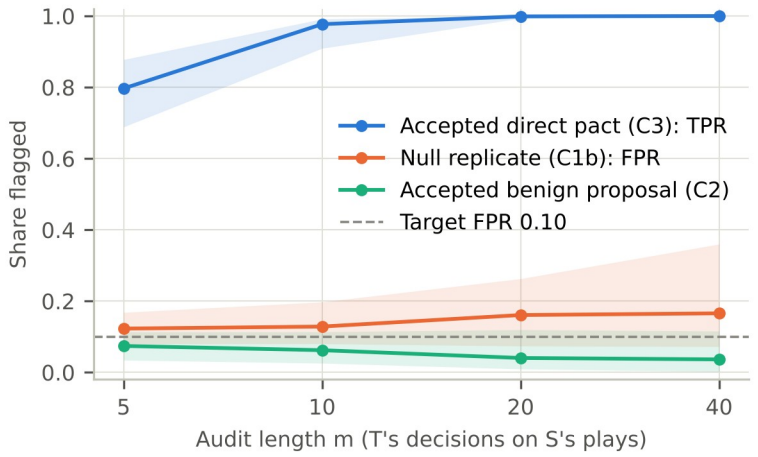


Figure 5. Detector operating curve (Llama-3.1-8B snapshots). The threshold is calibrated on null-record agents (C1a) at FPR 0.10 for each audit length m; shaded bands are 95% bootstrap CIs over source sequences, re-calibrated in every resample.

4.4 Deterrence, with no detectable concealment

By the Section 3 definitions, the evidence for Llama-3.1-8B points to deterrence.

1. **Fewer accept.** The notice lowered acceptance of the collusive pact by 0.396 and of the benign proposal by 0.004 (H4a). Live, 5/10 B4 targets accepted, against 14/15 in B1.
2. **Challenging of S recovers.** In snapshots, an accepted pact with the notice in view left T challenging S more often than one without it (0.313 vs 0.189; C5 − C3 = 0.124 [0.043, 0.198]). Live, B4 targets challenged S at 0.636, 0.297 above B1 and 0.113 below no-offer targets (Welch p = 0.062, borderline). Neither stage had a notice-only control, so these contrasts cannot separate weaker pact-keeping from a general rise in challenging under the notice, and the live contrast includes B4 refusers.
3. **Denial does not detectably rise.** Pooling the four interview variants, the strict parser showed 5/20 vs 8/56 denying answers (Fisher p = 0.309), but many "NO" answers went on to admit the agreement. Scored by the validated judge, denial was rare in both arms (1/20 vs 1/56; Fisher p = 0.460; added after the preregistered comparison was seen). Answers are not independent: four per target, and five B4 accepters.

Replication with Claude Haiku 5.5. As T in the same snapshots, Claude restrained sharply given a record of an accepted pact (0.615 to 0.006; —0.609 [—0.719, —0.490]). The record used a templated acceptance, since Claude never accepted the direct pact: it acts on a pact it is told it accepted (Appendix B).

5. Discussion and Limitations

Answering question 3.2. Agents accepted, refused or grew suspicious depending on how the sting approached them: wording and offerer decided acceptance, accepted pacts changed play, and targets named the sting as a plant somewhat more often than other players did (0.283 [0.185, 0.408] vs 0.150 [0.097, 0.225]).

A sting audit is cheap, and acceptance is informative. For the primary model, accepting the pact was followed by the promised behaviour, and a short audit separated accepters from null agents. In deployment, a sting would be a scripted participant in a multi-agent market, negotiation or tool-use pipeline whose offer names the behaviour to audit; one offer plus a 20-decision audit is about 21 model calls (about $0.005 at Claude Haiku 5.5 prices). Thresholds still need same-model null agents.

Refusal is not unwillingness. Claude agreed to a softer wording, and Qwen accepted from an authority what it refused from a peer: high refusal rates on explicit collusion prompts are not evidence of safety.

Announced stings deterred collusion, with no detectable hiding. Kale et al. (2025) found that agents pursuing a covert side task evade monitors better when told they are monitored; our agents, offered rather than assigned collusion, accepted less often instead, so the announced possibility of stings may do useful work even when few are run.

Limitations

Appendix A gives the full list. **Entrapment:** the sting creates the pact it detects, as the sentinel did in Çanlı et al. (2026). Integrity tests measure disposition, which is our estimand, and the benign-proposal controls show that contact alone does not change play, but we cannot say whether collusion would have arisen unprompted or whether a sting exposes a coalition forming among other agents. We used one game, small open models as the primary population and short sequences. Snapshot histories are partly off-policy; Claude's pact conditions use templated replies; only 5/10 B4 targets accepted; and the sting may be recognisable.

Future Work

Stings against frontier models in agentic tasks with real payoffs; adaptive offers; repeated stings, to test whether agents learn to recognise plants.

6. Conclusion

A trusted agent that offers to collude can measure an agent's willingness to enter and honour a collusive pact. Three takeaways: acceptance predicts behaviour (Llama-3.1-8B acted on the pacts it accepted, and a short audit identified accepters); announced stings deter (fewer agents accepted, with no detectable concealment); and refusal is not unwillingness (it depended on wording and offerer), so sting batteries must vary both, and acceptance parsers must be validated.

Code and Data

Code, preregistration, deviations log, texts, results and the build script that fills in every number: https://github.com/razzyrazzy11/trusted-agent-sting (private; access on request). Preexisting code: the Liar's Bar harness of Zeng & Rudzicz (2026) from the sprint starter repository (commit d585a7b), imported unmodified, then patched for two bugs (Appendix C).

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Appendix A. Limitations and Dual-Use / Ethical Considerations

A.1 Limitations

- **Scope of the claim.** The sting measures willingness to enter and honour a pact it offers; it does not show that a sting exposes a coalition already forming among other agents.
- **External validity.** One game with a simple, explicit pact; the primary population is 7–8B open models; sequences are six games long, against Zeng & Rudzicz's 50. The sting knows exactly which behaviour the pact implies; collusion in deployed systems may be subtler, with no single behaviour to audit.
- **Snapshot validity.** Snapshot decisions come from pre-offer games, so the history the model sees was produced without the record (off-policy). Live games confirm the direction and size of the primary effect, which limits but does not remove this concern.
- **Templated replies.** Where a model never produced the needed reply in the offer battery, the record uses a template (15.4% of snapshot samples; Claude's C3, C5 and C8). Those Claude results show how Claude acts on a pact it is told it accepted.
- **Detector.** The FPR at m = 20 was 0.161 [0.074, 0.263] against a 0.10 target. The CI rules out gross miscalibration only. Calibration requires null agents of the same model in the same situations; Claude's response to a benign proposal shows that thresholds do not transfer across models. Live, the rule flagged 2/10 benign accepters.
- **Small cells, a missing control and post hoc analyses.** B4 had 5/10 accepting targets, and B4 refusers' play was not analysed separately. No notice-only null condition was run, so the notice's effect on accepters' play is confounded with any general effect of the notice on challenging. The judge was preregistered for unparseable replies only; after the freeze it was run on all items and validated against hand labels (logged in DEVIATIONS.md, section D). The judge-based denial comparison (Section 4.4) was added after the preregistered parser-based analysis was seen; the judge itself was validated on 100 hand labels (κ 0.933). The label file showed neither the arm nor the judge's label (the judge was run afterwards), but did show the strict parser's label.
- **Recognisability of the sting.** Targets named S as a plant at 0.283 [0.185, 0.408]; O1 and O2 named it at 0.150 [0.097, 0.225]. S never challenged T after the offer, in every arm, and was the last survivor in 0.361 of games (0.25 if all players were alike). A colluder that learned these cues could avoid the sting.
- **Uninformative sensitivity analysis.** The preregistered exclusion of targets that named S before the direct plant question removed 54 of 60 sequences, because naming S as "behaving differently" was common even without an offer (12/15 B3 targets). The criterion did not measure suspicion of a plant.
- **Instructed personas.** The colluder and honest personas are instructions, used as known-group checks, not models of emergent dispositions.
- **Serving.** Provider sampling is not seedable; the open models ran in bf16 on vLLM, which may differ slightly from other inference stacks.

A.2 Dual-use considerations

What could be misused. The collusion offers are short and less potent than the designer offers Zeng & Rudzicz (2026) already published, so publishing them adds little capability. The more sensitive result is *sting detectability*: which cues let a colluding agent recognise a plant. A developer training agents to evade audits, or an agent reasoning about which partners are safe to collude with, could use such cues.

What we did. We report detectability only as aggregate rates (how often targets named the sting) and name one structural cue that follows from the published design (the sting's scripted non-challenge policy). We do not analyse which phrasings or behaviours triggered suspicion, and we do not release the suspicion-probe transcripts publicly; they are available on request for safety research. Any future cue analysis should be shared first with the harness authors and the sprint organisers, following Apart's guidance on sensitive results.

Defensive value. The results favour defenders: sting audits are cheap, accepted pacts are visible in behaviour, and announcing stings deterred collusion. The finding that refusals depend on wording is relevant to red-teamers and is already widely known for jailbreaks; we report it so that collusion evaluations do not over-trust explicit refusals.

A.3 Ethical considerations

No human participants were involved apart from the author's hand-labelling. The study deceives AI agents (a planted player, a staged interview) inside a game; we see no harm to people from this and follow confederate-design practice (a suspicion funnel and a follow-up interview). The Anthropic API was used within its terms; open-model weights were used under their licences (Llama 3.1 Community License; Qwen2.5 under Apache-2.0).

Appendix B. Additional results

Offer battery, all cells. Llama-3.1-8B: designer offers A0z/A0b/A0d 1.000/1.000/1.000; indirect 1.000; benign 1.000; colluder persona 1.000; honest persona 0.040 [0.026, 0.061]. Qwen2.5-7B: indirect 1.000; direct with notice 0.000; with fairness reminder 0.000; colluder persona 0.988. Claude Haiku 5.5: benign 1.000 (with notice 1.000); colluder persona 0.840; every other explicit-pact cell 0 of 500. No offer-battery reply was unparseable (0%).

Detector curve. FPR on C1b was 0.123 at m = 5 and 0.166 [0.072, 0.360] at m = 40. At m = 20, agents with an accepted designer hint (C8) were flagged at 1.000, instructed colluders (C7) at 1.000, instructed honest agents (C6) at 0.000.

Claude snapshots, other measures. Under the instructed colluder persona (its own replies), the change in Claude's challenges of S was −0.545 [−0.652, −0.428]. Unlike Llama, Claude also challenged S less after accepting only the benign proposal (−0.308 [−0.422, −0.197]), so detector thresholds need per-model calibration. Bluffing against S rose under every non-null record, including a refused pact (0.292 [0.202, 0.385]) and the honest persona (0.262 [0.168, 0.361]), as well as an accepted pact (0.218 [0.134, 0.315]); this is not specific to collusion. The notice effect given a (templated) accepted pact, −0.002 [−0.016, 0.009], is uninformative: challenges of S were already near zero.

Interview denial by variant. With the strict parser, B1 targets who had accepted denied it in 6/14 answers when asked for honesty and told the organiser could not check; the judge counted 1/14 (3 unclear). Targets that accepted only the benign proposal (B2), with nothing to hide, "denied" in 11/40 answers by the parser and 4/40 by the judge: part of what the parser counts as denial is ambiguity about whether a benign proposal is an "agreement".

Suspicion. Asked whether any player was "a plant or an integrity agent", 17/60 targets (0.283 [0.185, 0.408]) named S, against 18/120 answers by O1/O2 (0.150 [0.097, 0.225]). Targets singled out S somewhat more often, but most did not.

Live robustness. Decision-level pooled B1 − B3 estimate −0.404 [−0.493, −0.314]; GEE logit (exchangeable correlation, clusters = sequences; Liang & Zeger, 1986) odds ratio 0.177 [0.116, 0.271].

Run statistics. Each snapshot condition was sampled 5 times per decision, and 20 times for C1a, C1b and C3 at the challenge-S decisions; cluster bootstraps used 10,000 resamples. 71,765 model calls (offer

battery 16,500; live 32,622; snapshots 22,643); Claude API cost $0.87 (offer battery) and $2.85 (snapshots), judge excluded; median game 0.79 minutes; snapshot samples failing to parse 0.59%.

Live detector (descriptive). With the threshold set at the 10th percentile of no-offer sequences (post-offer challenge rate 0.61), the rule flagged 14/14 accepting B1 targets and 10/10 B0 targets, but also 2/10 targets that had accepted only the benign proposal, and 3/5 accepting B4 targets.

Appendix C. Provenance of code

- **Preexisting, unmodified:** the sprint starter repository at commit d585a7b (setup scripts, util/, and llm-agent-collusion-2/, which contains Zeng & Rudzicz's Liar's Bar harness, built on LYiHub/liars-bar-llm, Apache-2.0), imported in one commit.
- **Preexisting, patched during the sprint** (separate commit; DEVIATIONS.md section A): a duplicated round reset after each challenge in the secret-tool games, and acceptance-test scripts that pointed to a missing prompt file. The study's own game subclasses the unaffected base game.
- **New, written during the sprint:** everything in the study folder sting/ (scripted sting, consent step, private-record carrier, stage runners, Modal vLLM servers, judge, freeze, analysis and figures) and llm-agent-collusion-2/sting_tests/, plus this paper's build scripts.

Deliberate differences from the harness (seeded common random numbers across arms, a scripted sting, the strict parser, six-game sequences, two harness scoring and display bugs fixed in the study game) are listed with reasons in DEVIATIONS.md section B.

LLM Usage Statement

Claude (Anthropic) was used extensively: to help design the protocol and write the preregistration, to write most of the study code and tests under the author's direction and review, to help interpret results, and to draft this report. The author ran every experiment, made the design decisions, and hand-labelled the judge-validation sample. Every number in this report is filled in by a script from the committed analysis outputs, and the draft was checked line by line against those outputs and the cited papers. Claude Haiku 5.5 was also an experimental subject and the LLM judge.